

UNIT-3(Public Key Crypto System)

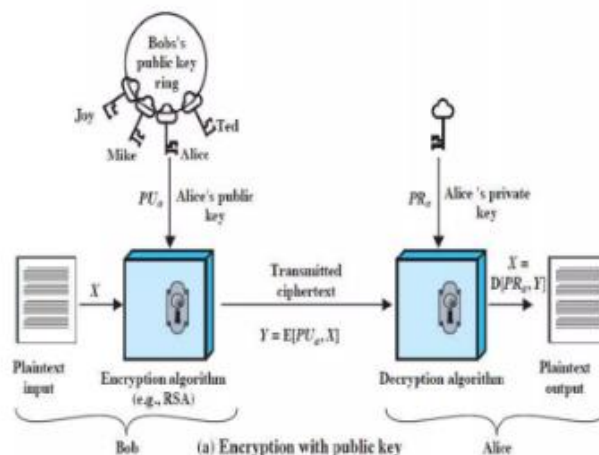
Public Key Crypto systems/Asymmetric Cryptosystems

A Public Key Crypto system is an encryption system that uses a pair of keys: a **public key** for encryption and a **private key** for decryption. The keys are mathematically related but computationally infeasible to derive one from the other. Public key cryptography is also called **asymmetric cryptography**.

A public key encryption scheme has following ingredients:

- ✚ **Plain text:** Readable message
- ✚ **Encryption Algorithm:** Perform operation on plain text
- ✚ **Public and Private Key:** This is a pair of key if one key is used for encryption; the other key is used for decryption.
- ✚ **Cipher text:** It is a scrambled /non-readable message produced as output.
- ✚ **Decryption Algorithm:** This algorithm accepts the cipher text and the matching key and produce original message.

Public-Key Cryptography

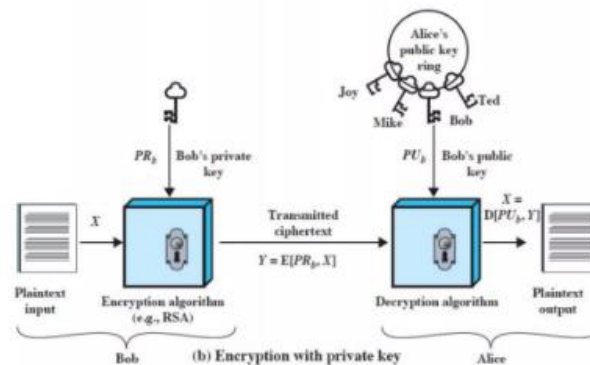


Key Characteristics:

- **Two keys:** Public (shared with everyone) and Private (kept secret).
- **One-way security:** Knowing the public key does not reveal the private key.
- Based on mathematical problems that are hard to solve (e.g., factoring large primes, discrete logarithm).
- Slower than symmetric encryption, often used with it in hybrid systems.

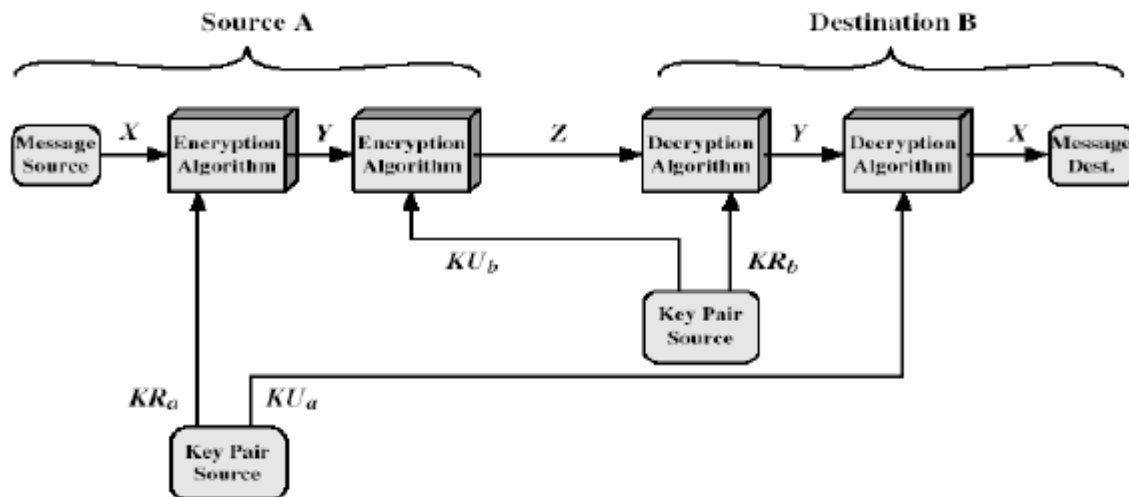
Public key Cryptography (Authentication)

Public-Key Cryptography



- In this case, Bob prepares a message to Alice and encrypt using Bob's private key before transmitting.
- Alice can decrypt the message using Bob's public key, because the message was encrypted using Bob's private key.
- Only Bob could have prepared the message, therefore the entire message serves as a digital signature.
 - $y = \text{EKRa}(X)$
 - $x = \text{DKUa}(y)$

Public key Cryptography (Authentication and Confidentiality)



- The process we seen before is only provide authentication.
- It does not provide message confidentiality.
- That is, the message being sent by sender but not safe from eavesdropping.
- In this case first we encrypt the message using A's private key than again encrypt the message using B's public key.
- At receiver side first b decrypt the message using B's private key and again decrypt the message using A's public key
- Thus confidentiality is provided.

How it Works:

1. The sender encrypts a message using the recipient's **public key**.
2. Only the recipient can decrypt it using their **private key**.
3. For digital signatures, the sender signs using their **private key**, and anyone can verify it using the **public key**.

Common Algorithms:

- RSA(Rivest–Shamir–Adleman)
- ElGamal
- Elliptic Curve Cryptography(ECC)
- Diffie–Hellman(for key exchange)

Applications:

1. **Secure Communication:** Encrypting data so only the intended recipient can read it.
2. **Digital Signatures:** Verifying the sender's identity and ensuring data integrity.
3. **Key Exchange:** Securely sharing symmetric keys over insecure channels (e.g., Diffie–Hellman).
4. **Authentication:** Verifying users or systems before granting access.
5. **SSL/TLS in HTTPS:** Securing web traffic between browsers and servers.
6. **Email Security:** Using PGP(Pretty Good Privacy) for encrypting and signing messages.

Advantages:

- No need to share private keys.
- Supports both encryption and authentication.

Disadvantages:

- Slower than symmetric cryptography.
- Requires careful key management.

Applications for public Key Cryptosystems

Secure Web Browsing (SSL/TLS): Encrypts traffic between your browser and websites via HTTPS, protecting login details and credit card numbers.

Digital Signatures: Verifies who sent a message and proves the file was not altered in transit by signing data with a private key and checking it with a public key.

Secure Remote Access (SSH): Allows IT administrators to log into servers securely without exposing passwords.

Crypto currency and Blockchain: Uses public keys as wallet addresses and private keys to sign and authorize transactions.

Encrypted Email (PGP/S-MIME): Scrambles email contents so that only the designated recipient with the

correct private key can read them.

Identity and Access Management: Powers authentication tools like WebAuthn and digital certificates to confirm user and server identity.

Requirements and Cryptanalysis

1. Requirements for a Good Cryptosystem

A secure crypto system must meet certain requirements to ensure confidentiality, integrity, and reliability of data.

a) Security Requirements:

- **Confidentiality:** The ciphertext should reveal nothing about the plaintext without the key.
- **Resistance to Cryptanalysis:** The system should with stand all known attacks (brute force, analytical attacks).
- **Key Secrecy:** The security should depend on keeping the key secret, not the algorithm.
- **Large Key Space:** The number of possible keys should be large enough to make brute-force attacks infeasible.
- **Randomness:** Keys and initialization vectors must be generated randomly to avoid patterns.
- **Efficiency:** The system should be fast enough for practical use without excessive computational overhead.
- **Error Handling:** Small errors in transmission should not reveal information about the plaintext.

2. Cryptanalysis

Crypt analysis is the study of analyzing cryptographic systems to find weaknesses and break encryption without knowing the secret key.

Types of Cryptanalysis Attacks:

1. **Cipher text-Only Attack(COA):**
 - The attacker only has access to cipher texts and tries to deduce the plaintext or key.
2. **Known-Plain text Attack (KPA):**
 - The attacker has both plaintext and its corresponding cipher text and attempts to find the key.
3. **Chosen-Plaintext Attack (CPA):**
 - The attacker can choose arbitrary plaintexts and obtain their cipher texts to find the key.
4. **Chosen-Ciphertext Attack (CCA):**
 - The attacker can choose cipher texts and obtain the corresponding plaintexts to find weaknesses.

5. **Side-Channel Attack:**

- Uses information leaked from the physical implementation (timing, power consumption, etc.).

6. **Brute Force Attack:**

- Tries all possible keys until the correct one is found; success depends on key size.

Goal of Cryptanalysis:

- To recover the secret key.
- To deduce the original plaintext without the key.
- To find weaknesses in the cryptographic algorithm or its implementation.
- RSA algorithm

RSA Algorithm

Definition:

RSA (Rivest–Shamir–Adleman) is a **public key cryptographic algorithm** used for secure data transmission. It is based on the mathematical difficulty of factoring large prime numbers. RSA supports both **encryption/decryption** and **digital signatures**.

Key Concepts:

- **Asymmetric encryption:** Uses two keys—Public key (for encryption) and Private key (for decryption).
- **Security basis:** Factoring the product of two large prime numbers is computationally infeasible.

Steps of RSA Algorithm:

Step-1: Select two prime numbers p and q where $p \neq q$.

Step-2: Calculate $n = p * q$.

Step-3: Calculate $\Phi(n) = (p-1) * (q-1)$.

Step-4: Select e such that, e is relatively prime to $\Phi(n)$, i.e. $(e, \Phi(n)) = 1$ and $1 < e < \Phi(n)$

Step-5: Calculate $d = e^{-1} \bmod \Phi(n)$ or $ed = 1 \bmod \Phi(n)$.

Step-6: Public key = $\{e, n\}$, private key = $\{d, n\}$.

Step-7: Find out cipher text using the formula,

$C = P^e \bmod n$ where, $P < n$ where C = Cipher text, P = Plaintext, e = Encryption key and n = block size.

Step-8: $P = C^d \bmod n$. Plaintext P can be obtain using the given formula. where d = decryption key

RSA algorithm explanation with example step by step

Step-1: Select two prime numbers p and q where $p \neq q$.

Example, Two prime numbers $p=13$, $q=11$.

Step-2: Calculate $n=p*q$.

Example, $n = p*q = 13*11 = 143$.

Step-3: Calculate $\Phi(n)=(p-1)*(q-1)$.

Example, $\Phi(n)=(13-1)*(11-1) = 12*10 = 120$.

Step-4: Select e such that, e is relatively prime to $\Phi(n)$, i.e. $(e, \Phi(n))=1$ and $1 < e < \Phi(n)$.

Example, Select $e=13$, $\gcd(13, 120)=1$.

Step-5: Calculate $d = e^{-1} \bmod \Phi(n)$ or $e*d \bmod \Phi(n) = 1$

Example, Finding $d: e*d \bmod \Phi(n) = 1$

$$13 * d \bmod 120 = 1$$

(How to find: $d * e = 1 \bmod \Phi(n)$)

$$d = ((\Phi(n)*i)+1)/e$$

$$d = (120+1)/13 = 9.30 (\because i=1)$$

$$d = (240+1)/13 = 18.53 (\because i=2)$$

$$d = (360+1)/13 = 27.76 (\because i=3)$$

$$d = (480+1)/13 = 37 (\because i=4)$$

Step-6: Public key = $\{e, n\}$, private key = $\{d, n\}$.

Example, Public key = $\{13, 143\}$ and private key = $\{37, 143\}$.

Step-7: Find out *cipher text* using the formula, $C = P^e \bmod n$ where, $P < n$.

Example, Plaintext $P=13$. (Where, $P < n$) $C = P^e \bmod n =$

$$13^{13} \bmod 143 = 52.$$

Step-8: $P = C^d \bmod n$. Plaintext P can be obtain using the given formula.

Example, Cipher text C=52

$$P = C^d \bmod n = 52^{37} \bmod 143 = 13.$$

Applications of RSA:

- Secure data transmission.
- Digital signatures.
- Key exchange insecure communication protocols (e.g., SSL/TLS).

Advantages

- **Security:** RSA algorithm is considered to be very secure and is widely used for secure data transmission.
- **Public-key cryptography:** RSA algorithm is a public-key cryptography algorithm, which means that it uses two different keys for encryption and decryption. The public key is used to encrypt the data, while the private key is used to decrypt the data.
- **Key exchange:** RSA algorithm can be used for secure key exchange, which means that two parties can exchange a secret key without actually sending the key over the network.
- **Digital signatures:** RSA algorithm can be used for digital signatures, which means that a sender can sign a message using their private key, and the receiver can verify the signature using the sender's public key.
- **Widely used:** Online banking, e-commerce, and secure communications are just a few fields and applications where the RSA algorithm is extensively developed.

Disadvantages

- **Slow processing speed:** RSA algorithm is slower than other encryption algorithms, especially when dealing with large amounts of data.
- **Large key size:** RSA algorithm requires large key sizes to be secure, which means that it requires more computational resources and storage space.
- **Vulnerability to side-channel attacks:** RSA algorithm is vulnerable to side-channel attacks, which means an attacker can use information leaked through side channels such as power consumption, electromagnetic radiation, and timing analysis to extract the private key.
- **Limited use in some applications:** RSA algorithm is not suitable for some applications, such as those that require constant encryption and decryption of large amounts of data, due to its slow processing speed.
- **Complexity:** The RSA algorithm is a sophisticated mathematical technique that some individuals may find challenging to comprehend and use.
- **Key Management:** The secure administration of the private key is necessary for the RSA algorithm, although in some cases this can be difficult.
- **Vulnerability to Quantum Computing:** Quantum computers have the ability to attack the RSA algorithm, potentially decrypting the data.

Computational Aspects and Security of RSA

1. Computational Aspects

- **Key Size:**
 - Security depends on the size of the modulus n .
 - Common sizes: 2048bits(recommended), 3072 or 4096bits for higher security.
- **Speed:**
 - RSA encryption and decryption involve modular exponentiation, which is computationally intensive.
 - Encryption is usually faster than decryption if a small public exponent e (like 65537) is used.
- **Key Generation:**
 - Generating large prime numbers p and q is time-consuming but done infrequently.
- **Data Size Limitation:**
 - The plaintext must be smaller than n .
 - For large messages, RSA is used to encrypt asymmetric key, which then encrypts the data (hybrid encryption).

2. Security of RSA

- **Mathematical Basis:**
 - Relies on the difficulty of factoring large integers.
 - No known efficient algorithm exists for factoring numbers of the size used in modern RSA.
- **Brute Force Resistance:**
 - With large key sizes, trying all possible keys is infeasible.
- **Potential Weaknesses:**
 - **Small Key Size:** Keys smaller than 1024 bits are vulnerable to factoring attacks.
 - **Poor Randomness:** Weak random number generation during key creation can lead to predictable primes.

Implementation Flaws: Side-channel attacks (timing, power analysis) can leak key information.

- **Chosen Cipher text Attacks:** If not padded properly, RSA is vulnerable (e.g., Bleichenbacher attack).
- **Counter measures:**
 - Use strong, random prime generation.
 - Apply secure padding schemes (e.g., OAEP).
 - Use sufficiently large keys (≥ 2048 bits).
 - Protect against side-channel attacks with constant-time operations.

Exercise - 1

Question: P and Q are two prime numbers. P=7, and Q=17. Take public key E=5. If plain text value is 6, then what will be cipher text value according to RSA algorithm? Again calculate plain text value from cipher text.

Solution:

1. Two prime numbers P=7, Q=17
2. $n = P * Q = 17 * 7 = 119$ **n = 119**
3. $\Phi(n) = (P-1) * (Q-1) = (17-1) * (7-1) = 16 * 6 = 96$ **$\Phi(n) = 96$**
4. Public key E = 5. **E = 5**
5. Calculate d = 77. $d = ((\Phi(n) * i) + 1) / e$ **d = 77**
 $d = ((96*1)+1) / 5 = 19.4$
 $d = ((96*2)+1) / 5 = 38.6$
 $d = ((96*3)+1) / 5 = 57.8$
 $d = ((96*4)+1) / 5 = 77$ (Stop finding d because getting integer value)
6. Public key = {e, n} = {5, 119}, private key = {d, n} = {77, 119}.
7. Plain text PT = 6, $CT = PT^E \bmod n = 6^5 \bmod 119 = 41$. **Cipher Text = 41**
8. Cipher text CT = 41, $PT = CT^d \bmod n = 41^{77} \bmod 119 = 6$. **Plain Text = 6**

Exercise - 2

Question: In a public key cryptosystem using RSA algorithm, user uses two prime numbers 5 and 7. He chooses 11 as Encryption key, find out decryption key. What will be the cipher text, if the plaintext is 2? Decrypt the cipher text, what will be the value of plain text?

Solution:

1. Two prime numbers $p = 5, q = 7$
2. $n = p * q = 5 * 7 = 35$ **$n = 35$**
3. $\Phi(n) = (p-1) * (q-1) = (5-1) * (7-1) = 4 * 6 = 24$ **$\Phi(n) = 24$**
4. Public key $e = 11$. **$e = 11$**
5. Calculate $d = 11$. $d = ((\Phi(n) * i) + 1) / e$ **$d = 11$**
6. Public key = $\{e, n\} = \{11, 35\}$, private key = $\{d, n\} = \{11, 35\}$.
7. Plain text $P = 2, C = P^e \bmod n = 2^{11} \bmod 35 = 18$. **Cipher Text = 18**
8. Cipher text $C = 18, P = C^d \bmod n = 18^{11} \bmod 35 = 2$. **Plain Text = 2**

Exercise - 3

Question: P and Q are two prime numbers. $P=17$, and $Q=11$. Take public key $E=7$. If plain text value is 5, then what will be cipher text value & private key value according to RSA algorithm? Again calculate plain text value from cipher text.

Solution:

1. Two prime numbers $p = 17, q = 11$
2. $n = p * q = 17 * 11 = 187$ **$n = 187$**
3. $\Phi(n) = (p-1) * (q-1) = (17-1) * (11-1) = 16 * 10 = 160$ **$\Phi(n) = 160$**
4. Public key $e = 7$. **$e = 7$**
5. Calculate $d = 23$. $d = ((\Phi(n) * i) + 1) / e$ **$d = 23$**
6. Public key = $\{e, n\} = \{7, 187\}$, private key = $\{d, n\} = \{23, 187\}$.
7. Plain text $P = 5, C = P^e \bmod n = 5^7 \bmod 187 = 146$. **Cipher Text = 146**
8. Cipher text $C = 146, P = C^d \bmod n = 146^{23} \bmod 187 = 5$. **Plain Text = 5**

Exercise - 4

Question: P and Q are two prime numbers. P=3, and Q=11. Take public key E=3. If original message is 00111011, then what will be cipher text value & private key value according to RSA algorithm? Again calculate plain text value from cipher text.

Solution:

1. Two prime numbers $p = 3, q = 11$
2. $n = p * q = 3 * 11 = 33$ **$n = 33$**
3. $\Phi(n) = (p-1) * (q-1) = (3-1) * (11-1) = 2 * 10 = 20$ **$\Phi(n) = 20$**
4. Public key $e = 3$. **$e = 3$**
5. Calculate $d = 7$. $d = ((\Phi(n) * i) + 1) / e$ **$d = 7$**
6. Public key = $\{e, n\} = \{3, 33\}$, private key = $\{d, n\} = \{7, 33\}$.
7. Plain text $P = (00111011)_2 = (59)_{10}$, $C = P^e \bmod n = 59^3 \bmod 33 = 20$. **Cipher Text = 20**
8. Cipher text $C = 20$, $P = C^d \bmod n = 20^7 \bmod 33 = 26$. **Plain Text = 26**

Exercise - 5

Question: P and Q are two prime numbers. P=13, and Q=17. Take public key E=19. If original message is 12, then what will be cipher text value & private key value according to RSA algorithm? Again calculate plain text value from cipher text.

Solution:

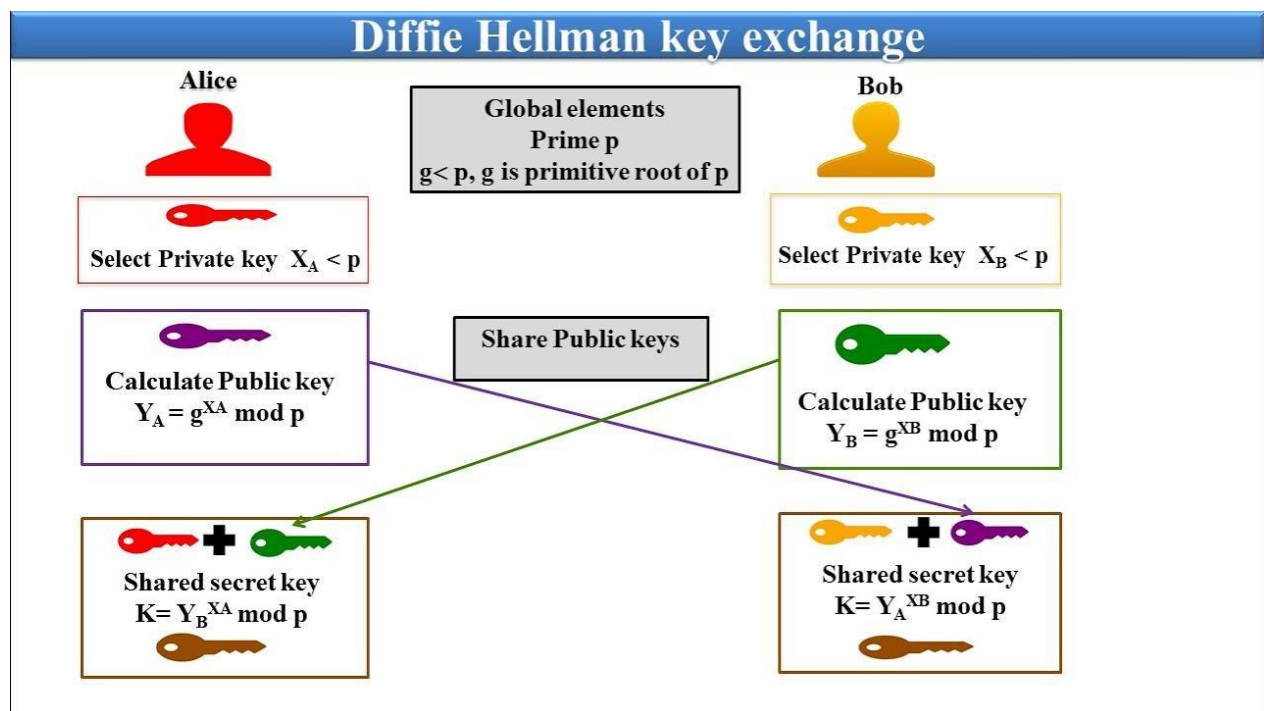
1. Two prime numbers $p = 13, q = 17$
2. $n = p * q = 13 * 17 = 221$ **$n = 221$**
3. $\Phi(n) = (p-1) * (q-1) = (13-1) * (17-1) = 12 * 16 = 192$ **$\Phi(n) = 192$**
4. Public key $e = 19$. **$e = 19$**
5. Calculate $d = 91$. $d = ((\Phi(n) * i) + 1) / e$ **$d = 91$**
6. Public key = $\{e, n\} = \{19, 221\}$, private key = $\{d, n\} = \{91, 221\}$.
7. Plain text $P = 12$, $C = P^e \bmod n = 12^{19} \bmod 221 = 181$. **Cipher Text = 181**
8. Cipher text $C = 181$, $P = C^d \bmod n = 181^{91} \bmod 221 = 12$. **Plain Text = 12**

Diffie – Hellman Key Exchange Algorithm

Definition:

Diffie–Hellman (DH) is a public key exchange protocol that allows two parties to establish a shared secret key over an insecure communication channel without sending the key itself. This shared key can then be used for symmetric encryption.

The Diffie Hellman algorithm widely known as Key exchange algorithm or key agreement algorithm developed by Whitefield Diffie and Martin Hellman in 1976. The purpose of the algorithm is to enable two users to securely exchange a key that can be used for subsequent encryption of messages. The algorithm itself is limited to the exchange of secret values.



Key Concepts:

- Based on the **discrete logarithm problem**, which is computationally hard to solve for large numbers.
- Involves a large prime number and a primitive root (generator).
- Both parties contribute to generating the shared key, but an eavesdropper cannot feasibly determine it.

Steps of the Diffie–Hellman Algorithm:

How it Works (Example with Sender and Receiver)

Step-1: Select q (prime number) and α (α is primitive root of q)

Step-2: User A Key Generation: select X_A , $X_A < q$

Calculate public key Y_A , $Y_A = \alpha^{X_A} \bmod q$

Y_A Shared with user B

Step-3: User B Key Generation: select X_B , $X_B < q$

Calculate public key Y_B , $Y_B = \alpha^{X_B} \bmod q$

Y_B shared with user A

Step-4: Calculation of secret key by user A: $K = (Y_B)^{X_A} \bmod q$

Step-5: Calculation of secret key by user B: $K = (Y_A)^{X_B} \bmod q$

Same Secret key generate both sides

$$K = (Y_B)^{X_A} \bmod q$$

$$= (\alpha^{X_B} \bmod q)^{X_A} \bmod q$$

$$= (\alpha^{X_B})^{X_A} \bmod q$$

$$= (\alpha^{X_A})^{X_B} \bmod q$$

$$= (\alpha^{X_A} \bmod q)^{X_B} \bmod q$$

$$= (Y_A)^{X_B} \bmod q$$

$$= K$$

Diffie–Hellman Key Exchange Algorithm explains with example:

Step–1: Select q (prime number) and α (α is primitive root of q)

Example, here $q = 7$, $\alpha = 17$.

Step–2: User A Key Generation: Select $X_A < q$,

Calculate **public key** Y_A and shared with user B: $Y_A = \alpha^{X_A} \bmod q$

Example, Here $X_A = 6$,

Calculate $Y_A = \alpha^{X_A} \bmod q \Rightarrow 17^6 \bmod 7 \Rightarrow 1$

Step – 3: User B Key Generation: Select $X_B < q$,

calculate **public key** Y_B and shared with user A: $Y_B = \alpha^{X_B} \bmod q$

Example, Here $X_B = 4$,

Calculate $Y_B = \alpha^{X_B} \bmod q \Rightarrow 17^4 \bmod 7 \Rightarrow 4$

Step – 4: Calculation of secret key by user A:

$K = (Y_B)^{X_A} \bmod q$

Example, Here $Y_B = 4$, $X_A = 6$

Calculate $K = (Y_B)^{X_A} \bmod q \Rightarrow 4^6 \bmod 7 \Rightarrow 1$

Step – 5: Calculation of secret key by user B:

$K = (Y_A)^{X_B} \bmod q$

Example, Here $Y_A = 1$, $X_B = 4$

Calculate $K = (Y_A)^{X_B} \bmod q \Rightarrow 1^4 \bmod 7 \Rightarrow 1$

Exercise - 1

Exercise – 1: $q = 353$, $\alpha = 3$, $X_A = 97$, $X_B = 233$.

Determine **Public Key** and **Shared key** for both users using Diffie-Hellman key exchange algorithm.

Solution:

1. Here $q = 353$, $\alpha = 3$

2. Calculate *Public key for user A*. $X_A = 97$,

$$Y_A = \alpha^{X_A} \bmod q = 3^{97} \bmod 353 = 40$$

$$Y_A = 40$$

3. Calculate *Public key for user B*. $X_B = 233$,

$$Y_B = \alpha^{X_B} \bmod q = 3^{233} \bmod 353 = 248$$

$$Y_B = 248$$

4. Calculation of *secret key by user A*,

$$K = (Y_B)^{X_A} \bmod q = 248^{97} \bmod 353 = 160$$

$$K = 160$$

5. Calculation of *secret key by user B*,

$$K = (Y_A)^{X_B} \bmod q = 40^{233} \bmod 353 = 160$$

$$K = 160$$

Exercise - 2

Exercise – 2: $q = 23$, $\alpha = 5$, $X_A = 6$, $X_B = 15$.

Determine **Public Key** and **Shared key** for both users using Diffie-Hellman key exchange algorithm.

Solution:

1. Here $q = 23$, $\alpha = 5$
2. Calculate *Public key for user A*. $X_A = 6$,
$$Y_A = \alpha^{X_A} \bmod q = 5^6 \bmod 23 = 8$$
3. Calculate *Public key for user B*. $X_B = 15$,
$$Y_B = \alpha^{X_B} \bmod q = 5^{15} \bmod 23 = 19$$
4. Calculation of *secret key by user A*,
$$K = (Y_B)^{X_A} \bmod q = 19^6 \bmod 23 = 2$$
5. Calculation of *secret key by user B*,
$$K = (Y_A)^{X_B} \bmod q = 8^{15} \bmod 23 = 2$$

$$Y_A = 8$$

$$Y_B = 19$$

$$K = 2$$

$$K = 2$$

Exercise - 3

Exercise – 3: $q = 11$, $\alpha = 17$, $X_A = 8$, $X_B = 12$.

Determine **Public Key** and **Shared key** for both users using Diffie-Hellman key exchange algorithm.

Solution:

1. Here $q = 11$, $\alpha = 17$
2. Calculate **Public key for user A**. $X_A = 8$,
$$Y_A = \alpha^{X_A} \bmod q = 17^8 \bmod 11 = 4$$
3. Calculate **Public key for user B**. $X_B = 12$,
$$Y_B = \alpha^{X_B} \bmod q = 17^{12} \bmod 11 = 3$$
4. Calculation of **secret key by user A**,
$$K = (Y_B)^{X_A} \bmod q = 3^8 \bmod 11 = 5$$
5. Calculation of **secret key by user B**,
$$K = (Y_A)^{X_B} \bmod q = 4^{12} \bmod 11 = 5$$

$$Y_A = 4$$

$$Y_B = 3$$

$$K = 5$$

$$K = 5$$

Man-in the middle attack:

- Man-in-the-middle attack (Meet-in-the-middle attack) is a common type of security attack that allows attackers to eavesdrop on the communication between two users.
- The attack takes place in between two users, allowing the attacker to “listen” to a conversation, without permission of users, hence the name “man-in-the-middle.”
- Here’s an analogy: Alice and Bob are having a conversation; Darth (Attacker) wants to eavesdrop on the conversation but also remain transparent.
- Darth could tell Alice that he was Bob and tell Bob that she was Alice.
- This would lead Alice to believe she’s speaking to Bob, while actually revealing her part of the conversation to Darth.
- Darth could then gather information from this, alter the response, and pass the message along to Bob (who thinks he’s talking to Alice).
- As a result, Darth is able to transparently hijack their conversation.

- Purpose of Diffie – Hellman key exchange algorithm is exchanging public keys and calculates secret keys using public keys.
- Then subsequent messages (further communications) are encrypted using secret keys.

How Man In The Middle of Network Works



How to prevent from Man-in-the-Middle Attack

- Strong WEP/WAP Encryption on Access Points
- Strong Router Login Credentials
- Virtual Private Network
- Use of HTTPS

Que: How can an attacker exploit Diffie-Hellman to perform a MITM attack? Illustrate with a diagram.

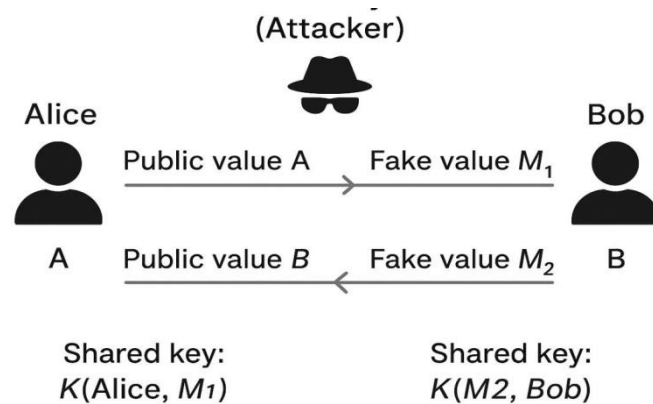
Answer:

In a **Man-in-the-Middle (MITM) attack** on the Diffie-Hellman key exchange, the attacker secretly intercepts and manipulates the communication between two parties (say Alice and Bob).

1. Alice sends her public value to Bob.
2. The Attacker intercepts it and sends his own public value to Bob instead.
3. Bob replies with his public value, but Attacker again intercepts and sends his own value to Alice.
4. As a result:
 - Alice thinks she is sharing a secret key with Bob, but she is actually sharing a secret key with Attacker.

- Bob thinks he is sharing a secret key with Alice, but he is actually sharing a different secret key with Attacker.
5. Attacker now holds two different shared keys (one with Alice and one with Bob). This allows Attacker to:
- Decrypt any message sent by Alice, read or modify it, then re-encrypt with the Bob-Attacker key and forward it.
 - Do the same for messages from Bob to Alice.

Thus, Attacker can fully control the conversation without Alice or Bob knowing



Symmetric vs Asymmetric Key Encryption

Symmetric Key Encryption	Asymmetric Key Encryption
It only requires a single key for both encryption and decryption.	It requires two keys, a public key and a private key, one to encrypt and the other to decrypt.
The size of cipher text is the same or smaller than the original plain text.	The size of cipher text is the same or larger than the original plain text.
The encryption process is very fast.	The encryption process is slow.
It is used when a large amount of data needs to be transferred.	It is used to transfer small amount of data.
It only provides confidentiality.	It provides confidentiality, authenticity, and non-repudiation.
The length of key used is 128 or 256 bits	The length of key used is 2048 or higher
In symmetric key encryption, resource utilization is	In asymmetric key encryption, resource utilization

Symmetric Key Encryption	Asymmetric Key Encryption
low compared to asymmetric key encryption.	is high.
It is efficient as it is used for handling large amount of data.	It is comparatively less efficient as it can handle a small amount of data.
Security is lower as only one key is used for both encryption and decryption purposes.	Security is higher as two keys are used, one for encryption and the other for decryption.
Mathematical representation: $P = D(K, E(K, P))$ where K: encryption and decryption key P: plain text D: Decryption E(K, P): Encryption of plain text using K	Mathematical representation: $P = D(K_d, E(K_e, P))$ Where Ke: encryption key Kd: decryption key D: Decryption E(Ke, P): Encryption of plain text using encryption key Ke. P --> plain text
Examples: 3DES, AES, DES and RC4	Examples: Diffie-Hellman, ECC, El Gamal, DSA and RSA